

Starting & Derieved Positions

- **Introduction**

- **Movement can be initiated from many positions.**
- **To perform any exercise or movement, the individual first attains any stable position.**
- **The position which is adapted by the patient to perform the exercises or movement and to stabilize the body is known as **Starting Position**.**
- **In simple words, we can also say that the position from which the movement starts is termed as the **Starting Position**.**
- **There are five fundamental starting positions :**
 1. **Standing**
 2. **Kneeling**
 3. **Sitting**
 4. **Lying**
 5. **Hanging.**

- **All the other positions are derived by changing and altering the position of hands, legs and trunk are known to be **Derived Position**.**
- **All exercises begin in any starting position and then the position are changed/altered as per the need, requirement and case of patient,**
- **This helps the patient to correct their balancing power and the muscle strength as well.**

▪ **USES**

1. **To increase stability**
2. **To increase relaxation**
3. **To perform particular exercise**
4. **For localizing the activity**
5. **For balancing**
6. **For gaining muscle strength**

STANDING

- **Whole body is supported or aligned by smaller BOS**
- **Most difficult position to adopt for a longer time because of the smaller BOS**
- **Body posture in Standing (Starting Position) :**
 - **Heels – Together in the same line, Toes apart**
 - **Knees – Together and straight**
 - **Hips – Extended and slightly laterally rotated**
 - **Pelvis – Balanced on femoral head**
 - **Spine – Stretched to max length**
 - **Head – Ears levelled, eyes looking forward**
 - **Shoulder – Levelled**
 - **Arms – Hanging loosely to side**
 - **Palm – Facing inwards toward body**

▪ **Derived positions of Standing:**

1. High Standing

- **Standing on any raised platform**
- **Used to perform**
 - **Upper and lower limb exercises**
 - **Hanging, swinging the arm and leg**

2. Walk Standing

- **One leg placed forward, and the base is increased.**
- **More stability in this position**
- **Posture**
 - **Hip – Flexion**
 - **Knee – Extension**
 - **Ankle – Mid Plantar Flexion**

2. Walk Standing

- **Used for**
 - **Starting position for jogging**
 - **Hip, knee, ankle and trunk muscle self-stretching**
 - **Relaxation from prolonged standing**

3. Stride Standing

- **Both feet kept apart laterally**
- **Gives lateral stability**
- **Posture**
 - **Hip abduction**
 - **Knee Extension**
- **Used for**
 - **Trunk, hip, knee exercise**
 - **Stretch hip adductor**

4. Half Standing

- **Standing on one leg, other leg is flexed totally,**
- **Most unstable position.**
- **So can't be maintained for long time**
- **Muscle work :**
 - **Adductor of opp side**
 - **Spinal flexor support the spine**
 - **Supporting leg muscle work very strongly**
- **Used for :**
 - **Leg swinging exercises**
 - **Stretching knee and hip muscles**
 - **Balancing the body on smaller base of support**

5. Toe Standing

- **Standing on Toes and the toe raise the body**
- **COG increases and BOS decreases**
- **Unstable position**
- **Muscle work :**
 - **Plantar flexor**
 - **Dorsi flexor**
 - **Toe extensors**
- **Used for :**
 - **Stretching plantar flexor, toe extensor & dorsiflexor**
 - **Balancing the body on smaller base of support**

❖ **Some other derived position of Standing are :**

1. **Step standing**
2. **Close standing**
3. **Foot and onfoot standing**
4. **Cross standing**
5. **Wind standing**
6. **Bend standing**
7. **Reach standing**
8. **Stretch standing**
9. **Yard standing**
10. **Cross yard standing**
11. **Heave standing**

SITTING

- The position while sitting on stool or chair is considered as the fundamental starting position.
- Thigh and legs are relaxed on the floor, Trunk is same as standing
- BOS more, COG lies down
- More stable and less muscle work
- **Body posture in Standing (Starting Position) :**
 - Heels – Together in the same line, Resting on floor
 - Knees – Together and Flexed 90 degree
 - Hips – Flexed at 90 degrees
 - Spine – Stretched to max length
 - Head – Ears levelled, eyes looking forward
 - Shoulder – Levelled

▪ **Derived positions of Sitting:**

1. **Cross Sitting**

- **Both legs crossed and weight is transferred through both hips.**
- **Posture :**
 - **Hips – Flexed and laterally rotated**
 - **Knees – Flexed**
- **Used For :**
 - **Breathing exercises and reduce respiratory distress**
 - **Hip adductor stretching**
 - **Neck and trunk exercises**

2. **Side Sitting**

- **Possible either on one side of hip**
- **For Rt sitting, the right leg same as cross sitting, Left hip flexed abducted medially rotated, knee flexed and kept in the side.**

2. Side Sitting

- **Used for :**
 - **Progression from prone to sitting in MAT Activity**
 - **Perform trunk exercises**

3. Crook Sitting

- **Knee and hip are flexed, feet are kept forward sitting on the floor**
- **Posture :**
 - **Hip – Flexed**
 - **Knee – Flexed**
 - **Ankle – Neutral**
- **Used for :**
 - **MAT Activity**
 - **Breathing Exercise**
 - **Ankle and Neck exercises**

4. Long Sitting

- **Hip flexed, knee extended resting on the floor. Spine will be erect supported with wall.**
- **Used for :**
 - **Hip rotational movements**
 - **Ankle and knee exercises**
 - **Trunk exercises**

❖ **Some other derived position of sitting are :**

- 1. Stoop sitting**
- 2. Ride sitting**
- 3. Foot sitting**

LYING

- **Lying on the floor in supine position**
- **Arms side of the body, trunk and legs straight**
- **Most stable due to broader BOS**
- **Minimal Muscle work**

- **Muscle work in Lying (Starting Position) :**
 - **Neck side flexors counterbalance to keep the head in neutral position.**
 - **Flexor of lumbar counteracts with spinal extensor to maintain lumbar lordosis.**
 - **Medial rotators of the hip keeps the hip in neutral position.**

- **Used for :**
 - **Strengthening**
 - **Stretching**
 - **Breathing exercises**

▪ **Derived positions of Lying:**

1. Prone lying

- **Anterior portion of body is turned towards couch,**
- **Difficult for respiratory patients and elder people**
- **Used for :**
 - **Spinal and Neck extensor exercises**
 - **Back massage**
 - **Some upper and lower limb exercises**

2. Half Lying

- **Upper portion (Trunk) is lifted at inclined and other same as supine.**
- **Used for :**
 - **Breathing ex and postural drainage**
 - **Frenkle ex.**
 - **Upper limb, Lower limb and trunk exercises**

3. Crook Lying

- **Hip, Knee flexed, feet kept apart and other same as supine.**
- **Used for :**
 - **Abdominal muscle strengthening**
 - **Pelvic bridging and half bridging exercises**

4. Side Lying

- **Lying on one side of body**
- **Upper limb which is under the body is flexed and kept below head**
- **Used for :**
 - **Upper limb and lower limb exercises**
 - **Postural drainage**

KNEELING

- Like fundamental position of standing but the weight is transmitted through knee,
- **Position in Kneeling (Starting Position) :**
 - Leg resting on floor
 - Ankle plantar flexion
 - Knee flexed
- **Used for :**
 - Upper limb and trunk exercises
 - MAT Activities
 - Balance and coordination training

▪ **Derived positions of Kneeling:**

1. **Half Kneeling**

- **One leg in kneeling position**
- **Other leg's Hip 90 degree flexed, Knee 90 degree flexed, Ankle neutral**
- **Used for :**
 - **Balance and coordination exercises**
 - **Trunk exercise**
 - **Neck exercises**
 - **Strengthening exercises**
 - **MAT Activities**
 - **Bending exercises**

2. Kneel sitting

- **Sitting on heel by flexing knee and hip**
- **Used for :**
 - **Balance and coordination exercises**
 - **Trunk exercise**
 - **Neck exercises**
 - **Strengthening exercises of knee and hip muscles**
 - **MAT Activities**

3. Prone kneeling (Quadruped Position)

- **Trunk inclined forward, stabilized by upper limb placing on the floor, Other as kneeling**
- **Posture :**
 - **Trunk forward bending**
 - **Shoulder flexed, elevated and medially rotated**
 - **Elbow, Wrist and Finger extended**

3. **Prone kneeling (Quadruped Position)**

- **Used for :**
 - **MAT Activity**
 - **Coordination and Strengthening exercises**
 - **Cat and Camel exercises**

HANGING

- **Total body suspended by grasping the horizontal bar with the hand.**
- **Position in Hanging (Starting Position) :**
 - **Shoulder flexed, abducted & medially rotated**
 - **Elbow extended**
 - **Fingers flexed and grasping the rod**
- **Muscle work**
 - **Abductor and medial rotator of shoulder**
 - **Elbow flexor**
 - **Hip adductor**
 - **Plantar flexor**
- **Used for**
 - **Whole body swinging**
 - **Upper body strengthening and body building**

▪ **Derived positions of Hanging:**

1. Half Hanging

- **Same as hanging but only one side upper limb is involved.**
- **Total body weight is carried by only one side upper limb**
- **Most advanced strengthening programme of upper extremity**
- **Helpful for body builders and gymnastic people**